Front. Chem. Sci. Eng. 2025, 19(10): 95
https://doi.org/10.1007/s11705-025-2566-x

VIEWS & COMMENTS

Regenerative engineering AI: a new paradigm for the future of tissue regeneration

Cato T. Laurencin (✉)[1,2,3,4,5], Taraje Whitfield[1,4], Chrysoula Argyrou[1,4], Fatemeh S. Hosseini[1,4]

1 Cato T. Laurencin Institute for Regenerative Engineering, University of Connecticut Health Center, Farmington, CT 06030, USA
2 Department of Biomedical Engineering, University of Connecticut, Storrs, CT 06269, USA
3 Department of Materials Science and Engineering, University of Connecticut, Storrs, CT 06269, USA
4 Department of Orthopaedic Surgery, University of Connecticut Health, Farmington, CT 06030, USA
5 Department of Chemical & Bimolecular Engineering, University of Connecticut, Storrs, CT 06269, USA

© Higher Education Press 2025

Abstract For over a decade, regenerative engineering has been defined as the convergence of advanced materials sciences, stem cell sciences, physics, developmental biology, and clinical translation for the regeneration of complex tissues. Recently, the field has made major strides because of new efforts made possible by the utilization of another growing field: artificial intelligence. However, there is currently no term to describe the use of artificial intelligence for regenerative engineering. Therefore, we hereby present a new term, "Regenerative Engineering AI", which cohesively describes the interweaving of artificial intelligence into the framework of regenerative engineering rather than using it merely as a tool. As the first to define the term, regenerative engineering AI is the interdisciplinary integration of artificial intelligence and machine learning within the fundamental core of regenerative engineering to advance its principles and goals. It represents the subsequent synergetic relationship between the two that allow for multiplex solutions toward human limb regeneration in a manner different from individual fields and artificial intelligence alone. Establishing such a term creates a unique and unified space to consolidate the work of growing fields into one coherent discipline under a common goal and language, fostering interdisciplinary collaboration and promoting focused research and innovation.

1 Introduction

1.1 Overview of regenerative engineering

The progressive advancements in tissue regeneration that took place during the late 20th and the dawn of the 21st century were the main drives of Dr. Cato T. Laurencin who termed regenerative engineering as the convergence of advanced materials sciences, stem cell sciences, physics, developmental biology, and clinical translation for the regeneration of complex tissues [1−16]. Integrating these rapidly developing fields provided a paradigm shift in our understanding of the major constituents to be harnessed for effective tissue regeneration.

At the backbone of these principles lies the ability of advanced biomaterials to guide stem cell differentiation towards multiple lineages at a nano-scale level, according to the needs of the host tissue. This cell-matrix interplay is largely affected by the physiochemical characteristics, 3-dimensional structure, and surface topography of the materials that direct the mechanical and biological properties of the matrix leading to enhanced cell adhesion, proliferation and differentiation [17−24]. Furthermore, crucial insights from developmental biology serve to bring the field one step closer to successful limb regeneration by elucidating the underlying molecular mechanisms through the identification of important signaling pathways, small bioactive molecules, and growth factors that can in turn be incorporated into novel regenerative engineering approaches [25−34]. Ultimately, regenerative engineering represents a constantly evolving interdisciplinary framework for successful clinical translation with the capacity to encompass contemporary advancements from various scientific and technological sectors, including nanotechnology and artificial intelligence (AI) [2].

1.2 The rise of artificial intelligence

The umbrella term, AI, describes systems that can respond to various inputs to provide the output of

Received February 19, 2025; accepted April 1, 2025; online May 20, 2025

E-mail: laurencin@uchc.edu

intelligently completing a desired task such as using data to learn and make predictions (e.g., machine learning and deep learning) [35]. It is a relatively new and emerging tool that has led to the rapid advancement of many fields. Since its creation, however, AI has not only helped evolve various other fields, but AI itself has also evolved into various forms. Two such derivatives of traditional AI are generative and regenerative AI. Generative AI refers to the form of AI that takes inputs such as a text prompt, then utilizes generative models to produce new text, images, or other solicited outputs [35,36]. On the other hand, Regenerative AI is a newer adaptation. While Regenerative AI is also capable of generating desired outcomes, it can furthermore repair and improve its own functionality overtime as well as other existing media types [37−39]. Traditional AI systems are primarily used to analyze data and make predictions, while generative and regenerative AI go a step further by creating or modifying data. As part of a larger system, AI in all its forms, is currently being used to advance various fields by easily and efficiently performing difficult tasks that would otherwise take large amounts of time and energy [40]. Consequently, AI has revolutionized areas such as finance, entertainment, marketing, product design, medicine, and notably, regenerative engineering.

2 Regenerative engineering AI

The fundamental principles of regenerative engineering entail the convergence of comprehensive theories across five major multidisciplinary fields for regenerating damaged tissues and organs. Recently, the field has made major strides because of new efforts made possible by the utilization of another growing field: AI [41,42]. After over a decade, the use of AI in regenerative engineering has been established, but there is currently no term to describe the use of AI for regenerative engineering. Therefore, we hereby present a new term, “regenerative engineering AI”, to describe such work (Fig. 1).

2.1 Definition

As the first group to utilize this term, regenerative engineering AI is the interdisciplinary integration of artificial intelligence and machine learning within the

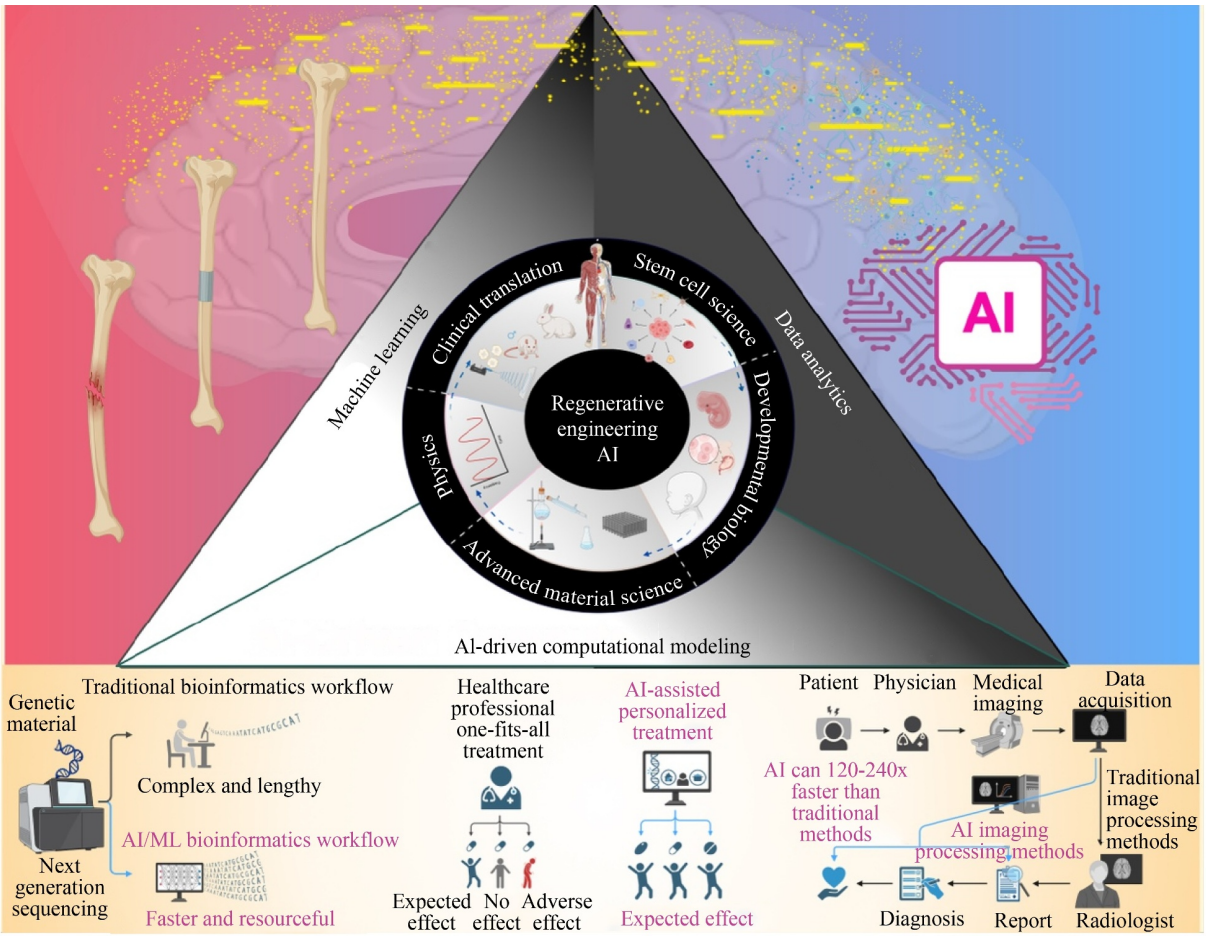

Fig. 1 Regenerative engineering AI describes the integration of artificial intelligence into the foundations of regenerative engineering.

fundamental core of regenerative engineering to advance its principles and goals (Fig. 1). It represents the synergetic relationship that arises from the convergence of AI-driven computational modeling, data analytics, and machine learning with advanced materials science, stem cell research, developmental biology, and clinical translation.

Regenerative engineering AI aims to optimize the design and development of complex tissue and organ systems by enabling precise predictions, innovative biomaterial discovery, enhanced control of cellular behavior, predictive modeling, optimal scaffold designs, simulation of biological processes, and analysis of large datasets to uncover new insights and efficient pathways for clinical application.

What makes regenerative engineering AI different from the mere use of AI in areas such as biomedical sciences is its focus on simultaneously converging the integration of AI in each subfield of regenerative engineering to provide novel innovations and solutions to complex problems, such as multi-tissue repair, which could not be achieved with AI and any individual field alone. While AI in individual fields may address specific, targeted problems, the term regenerative engineering AI covers the emerging multiplex solutions and techniques that integrate and optimize interactions and feedback loops between cells, scaffolds, forces, system-level models, physiological processes, simulations/predictions, and clinical translation. Additionally, regenerative engineering AI describes not only the ways in which AI advances specific disciplines, but also how data from disciplines such as stem cell science can lead to more accurate AI models that, in turn, further shape experimentation and innovation to advance the goals of regenerative engineering. This synergetic relationship where AI is not merely a tool, but an essential component interwoven into its framework, necessitates the generation of an all-encompassing and adaptable term dedicated to the new conceptual, methodical, and innovative ideologies that may arise from the co-evolution of AI and other fields as the pursuit of regenerating complex tissue systems continues—hence, the need for "regenerative engineering AI" as both a term and a paradigm.

2.2 Application of artificial intelligence in regenerative engineering

Regenerative engineering AI synthesizes and models data from diverse transdisciplinary principles and theories to generate comprehensive, holistic conclusions [43]. Despite its early stage, application of AI for regenerative engineering has been well-established, serving as a practical toolbox to cohesively create the consilience across multidisciplinary fields of regenerative engineering. Integration of AI-driven theories in tandem with pioneering technologies in stem cell sciences, advanced materials sciences, physics, developmental biology, and clinical translation harness AI's capability to enhance the regeneration efficacy [44].

One of the most challenging obstacles to overcome regarding successful tissue regeneration is our ability to accurately predict and model tissue growth. AI techniques are designed to process large volumes of data from various sources, and through advanced pattern recognition algorithms, it can provide predictive models that are able to adapt to the observed data. For example, a recent paper investigated the use of a Bayesian optimization network to reverse engineer the epithelial morphogenesis of Drosophila's wing disc [45]. This physics-based model aimed to mimic cell movement during development. The major advantage of the use of this technique was the ability to simultaneously process multiple factors that affect morphogenesis, like the numerous proteins that regulate shape changes and the diverse non-linear effects that they have on tissue growth. Also, the use of machine learning algorithms allowed the automatic adjustment of the model's parameters in a way that they matched the observed data, providing more realistic tissue modeling [45].

Another key advantage of the use of AI models in regenerative engineering is the acceleration of material discovery. In tandem with current developmental biology knowledge, regenerative engineering AI predicts the interplay of cells' behavior, fate, and their surrounding environment in response to advanced biomaterials to optimize biomaterial chemistry and formulation. As such, this enables scientists to enhance matrix design to ensure that engineered matrices not only replicate the hierarchical structure of damaged tissues but also actively guide cellular behavior towards the desired path [46]. Indeed, AI algorithms leverage the existing knowledge of body's natural regenerative potentials and limitations for prediction and theory modeling of the fate of regenerative strategies for ideal outcome. The Materials Genome Initiative is a great example of AI's ability to transform our approach in current research for novel biomaterials [47]. Some of the AI techniques that are employed in this project are the Recursive Feature Elimination, used to narrow down the necessary components of the material's fingerprint to predict a particular property, as well as the Gaussian Process Regression, designed to provide not only an output value, but also a quantitative uncertainty variance, leading to high-confidence predictions. This effort provides valuable insights to further optimize biomaterial synthesis and matrix fabrication strategies to engineer tissue-equivalent functional matrices and bypass the hurdle of precisely mimicking hierarchical architecture of damaged tissue for guided stem cell response [48]. Additionally, regenerative engineering AI provides atomic level insights about the biomaterial properties to

improve its mechanical characteristics, degradation rates, and biocompatibility [49]. By considering the unique structure of each tissue, the AI-refined biomaterials enable the fabrication of matrices with advanced structure tailored for specific tissue engineering needs, thus reducing trial-and-error while accelerating clinical translation.

Regenerative engineering AI provides comprehensive conclusions by cohesively merging and modeling data sets from across all the transdisciplinary principles and theories to implement holistic conclusions [50]. This method enables us to transcend the fragmented knowledge and skills of various disciplines, propelling us toward authentic transdisciplinary science. Figure 2 provides a detailed workflow of Regenerative Engineering AI, starting by data acquisition from various sources, including in vitro, animal and clinical studies. Subsequently, the large volume of data needs to be categorized and important features need to be extracted in order to be converted into measurable data that AI algorithms can process. These algorithms aim to provide accurate simulations, as well as optimization processes, depending on the context. The new AI-driven techniques or materials can then be tested, and the results can be compared with the predictions provided by the AI models. The observed data serve as input and can further be used to fine-tune the AI algorithms and provide new simulations, closer to reality. This workflow can be used in the context of a single tissue regeneration, however its power is evident in the attempt to regenerate multiple tissues, where there is an immense number of parameters defining the output with many interactions, often hidden among the various studies.

3 Conclusions

The computational and problem-solving capabilities of AI are already considerable and will continue to grow at an unprecedented pace over the next years alongside regenerative engineering techniques. By integrating AI into the core of regenerative engineering rather than using it as an external tool, regenerative engineering AI creates a unique and unified space for the field to gather and interpret large amounts of research data under the scope of complex tissue regeneration and, even more importantly, the capacity to predict and simulate innovative solutions with high precision and accuracy faster than ever before. Optimization of biomaterials design and properties, in silico modeling and big data analytics, are some of the other areas that regenerative engineering AI will focus on in the future alongside the advancements of the existing components of regenerative engineering. Regenerative engineering AI, therefore, consolidates the work of these growing fields into one coherent discipline, establishing a common language that researchers can use for interdisciplinary collaboration and focused research, thus pushing the field closer to successful human limb regeneration.

Acknowledgements This work was supported by funding from NIH T32 AR079114. We would like to thank Dr. Debolina Ghosh for manuscript editing and helpful discussions. All schematic illustrations were created using BioRender.

Competing interests The authors declare that they have no competing interests.

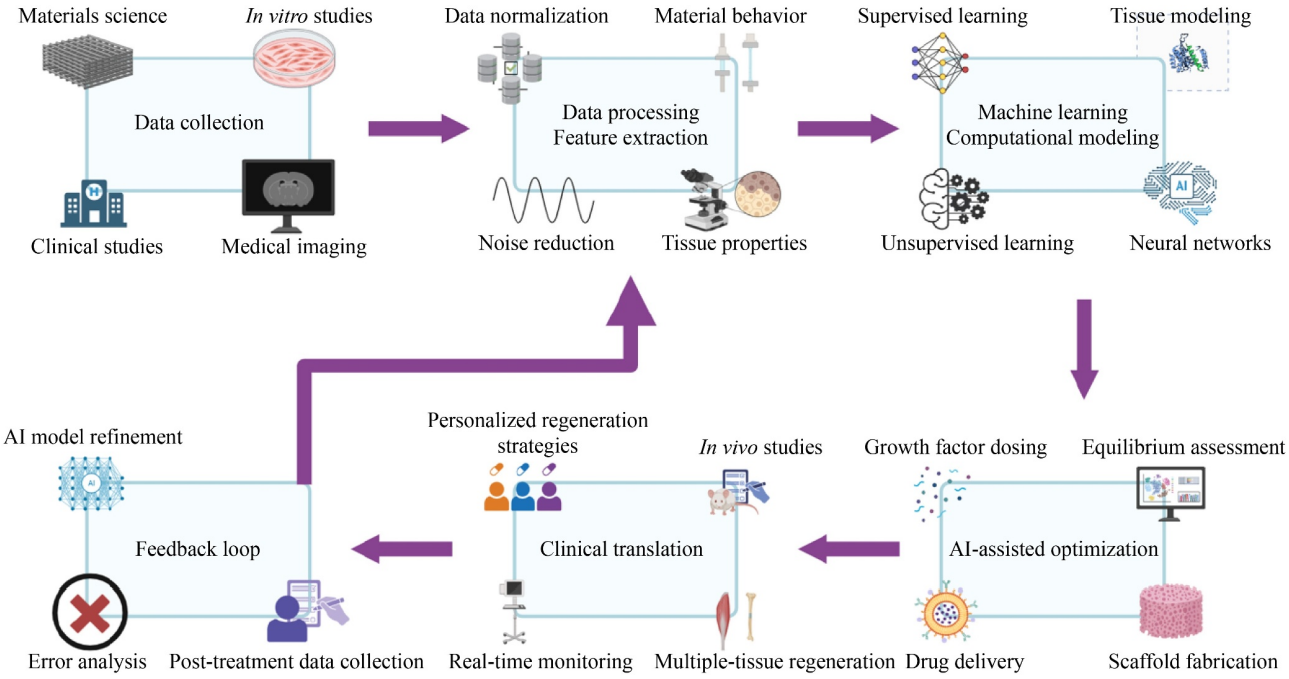

Fig. 2 Workflow diagram of regenerative engineering AI.

References

1. Laurencin C T, Khan Y. Regenerative engineering. Science Translation Medicine, 2012, 4(160): 160ed9
2. Laurencin C T, Hosseini F S, Daneshmandi L. Regenerative engineering: from convergence to consilience. Regenerative Engineering and Translational Medicine, 2024, in press
3. Abedini A A, Hosseini F, Laurencin C T. Regenerative engineering of a limb: from amputation to regeneration. Regenerative Engineering and Translational Medicine, 2023, 10(4): 461–479
4. Esdaille C J, Washington K S, Laurencin C T. Regenerative engineering: a review of recent advances and future directions. Regenerative Medicine, 2021, 16(5): 495–512
5. Shah S, Esdaille C J, Bhattacharjee M, Kan H M, Laurencin C T. The synthetic artificial stem cell (SASC): shifting the paradigm of cell therapy in regenerative engineering. Proceedings of the National Academy of Sciences of the United States of America, 2022, 119(2): e2116865118
6. Barui S, Ghosh D, Laurencin C T. Osteochondral regenerative engineering: challenges, state-of-the-art and translational perspectives. Regenerative Biomaterials, 2023, 10: rbac109
7. Tang X, Daneshmandi L, Awale G, Nair L S, Laurencin C T. Skeletal muscle regenerative engineering. Regenerative Engineering and Translational Medicine, 2019, 5(3): 233–251
8. Barajaa M A, Ghosh D, Laurencin C T. Decellularized extracellular matrix-derived hydrogels: a powerful class of biomaterials for skeletal muscle regenerative engineering applications. Regenerative Engineering and Translational Medicine, 2023, 11(1): 39–63
9. Hosseini F S, Laurencin C T. Advanced electrospun nanofibrous stem cell niche for bone regenerative engineering. Regenerative Engineering and Translational Medicine, 2022, 9(2): 165–180
10. Hosseini F S, Abedini A A, Chen F, Whitfield T, Ude C C, Laurencin C T. Oxygen-generating biomaterials for translational bone regenerative engineering. ACS Applied Materials & Interfaces, 2023
11. Barajaa M A, Otsuka T, Ghosh D, Kan H M, Laurencin C T. Development of porcine skeletal muscle extracellular matrix-derived hydrogels with improved properties and low immunogenicity. Proceedings of the National Academy of Sciences of the United States of America, 2024, 121(19): e2322822121
12. Esdaille C J, Ude C C, Laurencin C T. Regenerative engineering animal models for knee osteoarthritis. Regenerative Engineering and Translational Medicine, 2022, 8(2): 284–297
13. Ude C C, Shah S, Ogueri K S, Nair L S, Laurencin C T. Stromal vascular fraction for osteoarthritis of the knee regenerative engineering. Regenerative Engineering and Translational Medicine, 2022, 8(2): 210–224
14. Hosseini F S, Nair L S, Laurencin C T. Inductive materials for regenerative engineering. Journal of Dental Research, 2021, 100(10): 1011–1019
15. Hosseini F S, Laurencin C T. Advanced graphene ceramics and their future in bone regenerative engineering. International Journal of Applied Ceramic Technology, 2022, 19(2): 893–905
16. Vinikoor T, Dzidotor G K, Le T T, Liu Y, Kan H M, Barui S, Chorsi M T, Curry E J, Reinhardt E, Wang H, et al. Injectable and biodegradable piezoelectric hydrogel for osteoarthritis treatment. Nature Communications, 2023, 14(1): 6257
17. Otsuka T, Kan H M, Mengsteab P Y, Tyson B, Laurencin C T. Fibroblast growth factor 8b (FGF-8b) enhances myogenesis and inhibits adipogenesis in rotator cuff muscle cell populations in vitro. Proceedings of the National Academy of Sciences of the United States of America, 2024, 121(1): e2314585121
18. Arnold A M, Holt B D, Daneshmandi L, Laurencin C T, Sydlik S A. Phosphate graphene as an intrinsically osteoinductive scaffold for stem cell-driven bone regeneration. Proceedings of the National Academy of Sciences of the United States of America, 2019, 116(11): 4855–4860
19. Ogueri K S, Escobar Ivirico J L, Li Z, Blumenfield R H, Allcock H R, Laurencin C T. Synthesis, physicochemical analysis, and side group optimization of degradable dipeptide-based polyphosphazenes as potential regenerative biomaterials. ACS Applied Polymer Materials, 2019, 1(6): 1568–1578
20. Li W J, Laurencin C T, Caterson E J, Tuan R S, Ko F K. Electrospun nanofibrous structure: a novel scaffold for tissue engineering. Journal of Biomedical Materials Research, 2002, 60(4): 613–21
21. Mirdamadi E S, Attari S, Laurencin C T. Standardizing clinical trials for synthetic matrices in rotator cuff tears. Nature Reviews Bioengineering, 2025, in press
22. Ambrosio A M, Sahota J S, Khan Y, Laurencin C T. A novel amorphous calcium phosphate polymer ceramic for bone repair: I. Synthesis and characterization. Journal of Biomedical Materials Research, 2001, 58(3): 295–301
23. Borden M, Attawia M, Khan Y, El-Amin S F, Laurencin C T. Tissue-engineered bone formation *in vivo* using a novel sintered polymeric microsphere matrix. Journal of Bone and Joint Surgery. British Volume, 2004, 86(8): 1200–1208
24. Yu X, Botchwey E A, Levine E M, Pollack S R, Laurencin C T. Bioreactor-based bone tissue engineering: the influence of dynamic flow on osteoblast phenotypic expression and matrix mineralization. Proceedings of the National Academy of Sciences of the United States of America, 2004, 101(31): 11203–11208
25. Laurencin C T, Nair L S. The quest toward limb regeneration: a regenerative engineering approach. Regenerative Biomaterials, 2016, 3(2): 123–125
26. Cushnie E K, Ulery B D, Nelson S J, Deng M, Sethuraman S, Doty S B, Lo K W, Khan Y M, Laurencin C T. Simple signaling molecules for inductive bone regenerative engineering. PLoS One, 2014, 9(7): e101627
27. Otsuka T, Phan A Q, Laurencin C T, Esko J D, Bryant S V, Gardiner D M. Identification of heparan-sulfate rich cells in the loose connective tissues of the axolotl (ambystoma mexicanum) with the potential to mediate growth factor signaling during regeneration. Regenerative Engineering and Translational Medicine, 2020, 6(1): 7–17
28. Awale G M, Barajaa M A, Kan H M, Seyedsalehi A, Nam G H, Hosseini F S, Ude C C, Schmidt T A, Lo K W, Laurencin C T. Regenerative engineering of long bones using the small molecule

forskolin. Proceedings of the National Academy of Sciences of the United States of America, 2023, 120(22): e2219756120
29. Merriman Jr M A, Chapman J H, Whitfield T, Hosseini F, Ghosh D, Laurencin C T. Fat expansion not fat infiltration of muscle post rotator cuff tendon tears of the shoulder: regenerative engineering implications. Regenerative Engineering and Translational Medicine, 2023, 11(1): 1–14
30. Shah S, Ghosh D, Otsuka T, Laurencin C T. Classes of stem cells: from biology to engineering. Regenerative Engineering and Translational Medicine, 2024, 10(3): 309–322
31. Bhattacharjee M, Escobar Ivirico J L, Kan H M, Shah S, Otsuka T, Bordett R, Barajaa M, Nagiah N, Pandey R, Nair L S, et al. Injectable amnion hydrogel-mediated delivery of adipose-derived stem cells for osteoarthritis treatment. Proceedings of the National Academy of Sciences of the United States of America, 2022, 119(4)
32. Chapman J H, Ghosh D, Attari S, Ude C C, Laurencin C T. Animal models of osteoarthritis: updated models and outcome measures 2016–2023. Regenerative Engineering and Translational Medicine, 2024, 10(2): 127–146
33. Dzidotor G K, Moorhead Jr J B, Ude C C, Ogueri K S, Ghosh D, Laurencin C T. Functions and effectiveness of unloader, patellofemoral, and knee sleeve orthoses: a review. Regenerative Engineering and Translational Medicine, 2024, 10(2): 147–171
34. Amini A R, Laurencin C T, Nukavarapu S P. Bone tissue engineering: recent advances and challenges. Critical Reviews in Biomedical Engineering, 2012, 40(5): 363–408
35. Dimitrieska S. Generative artificial intelligence and advertising. Trends in Economics, Finance and Management Journal, 2024, 6(1): 23–34
36. Banh L, Strobel G. Generative artificial intelligence. Electronic Markets, 2023, 33(1): 63
37. Bhattacharya S, Saleem S M, Singh A, Singh S, Tripathi S. Empowering precision medicine: regenerative AI in breast cancer. Frontiers in Oncology, 2024, 14: 1465720
38. Varghese D J. Metaverse Liability and the Future of Tort Law. Available at SSRN 5043578, 2024
39. Ramamritham K. Machine learning: principles and practices. CSI Transactions on ICT, 2023, 11(4): 177–178
40. Shin Y, Kim S, Lee Y H. AI musculoskeletal clinical applications: how can AI increase my day-to-day efficiency? Skeletal Radiology, 2022, 51(2): 293–304
41. Gharibshahian M, Torkashvand M, Bavisi M, Aldaghi N, Alizadeh A. Recent advances in artificial intelligent strategies for tissue engineering and regenerative medicine. Skin Research and Technology, 2024, 30(9): e70016
42. Nosrati H, Nosrati M. Artificial intelligence in regenerative medicine: applications and implications. Biomimetics, 2023, 8(5): 442
43. Dzobo K, Adotey S, Thomford N E, Dzobo W. Integrating artificial and human intelligence: a partnership for responsible innovation in biomedical engineering and medicine. OMICS, 2020, 24(5): 247–263
44. Thimukonda Jegadeesan, Baldia J M, Basu B. Next-generation personalized cranioplasty treatment. Acta Biomaterialia, 2022, 154: 63–82
45. Kumar N, Mim M S, Dowling A, Zartman J J. Reverse engineering morphogenesis through Bayesian optimization of physics-based models. NPJ Systems Biology and Applications, 2024, 10(1): 49
46. Al-Kharusi G, Dunne N J, Little S, Levingstone T J. The role of machine learning and design of experiments in the advancement of biomaterial and tissue engineering research. Bioengineering, 2022, 9(10): 561
47. Kim C, Chandrasekaran A, Huan T D, Das D, Ramprasad R. Polymer genome: a data-powered polymer informatics platform for property predictions. The Journal of Physical Chemistry C, 2018, 122(31): 17575–17585
48. Giannitelli S M, Accoto D, Trombetta M, Rainer A. Current trends in the design of scaffolds for computer-aided tissue engineering. Acta Biomaterialia, 2014, 10(2): 580–94
49. Huang J S, Liew J X, Ademiloye A S, Liew K M. Artificial intelligence in materials modeling and design. Archives of Computational Methods in Engineering, 2020, 28(5): 3399–3413
50. Jayatilake S, Ganegoda G U. Involvement of machine learning tools in healthcare decision making. Journal of Healthcare Engineering, 2021, 2021: 6679512